Ferroelectricity in Hafnia Controlled *via* Surface Electrochemical State

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Abstract:

Ferroelectricity in binary oxides including hafnia and zirconia have riveted the attention of the scientific community due to highly unconventional physical mechanisms and the potential for integration of these materials into semiconductor workflows. Over the last decade, it has been

argued that behaviors such as wake-up phenomena and an extreme sensitivity to electrode and processing conditions suggests that ferroelectricity in these materials is strongly influenced by other factors, including electrochemical boundary conditions and strain. Here we argue that the properties of these materials emerge due to the interplay between the bulk competition between ferroelectric and structural instabilities, similar to that in classical antiferroelectrics, coupled with non-local screening mediated by the finite density of states at surfaces and internal interfaces. Via decoupling of electrochemical and electrostatic controls realized via environmental and ultra-high vacuum PFM, we show that these materials demonstrate a rich spectrum of ferroic behaviors including partial pressure- and temperature-induced transitions between FE and AFE behaviors. These behaviors are consistent with an antiferroionic model and suggest strategies for hafnia-based device optimization.

Main

The discovery of ferroelectricity in hafnia took the scientific community by surprise.[1, 2] These findings represented the first reliable observation of ferroelectricity in binary oxides, suggesting a mechanism very different from that in classical ABO_3 perovskites. Importantly, hafnia and related binary oxides allow for integration with semiconductor manufacturing workflows, a long-standing challenge for the ferroelectric community spurred by potential applications in non-volatile random access memories,[3, 4] tunneling barriers,[5, 6] field-effect transistors,[7, 8] and multiferroic devices.[9, 10] It is notable that initial reports of ferroelectricity in hafnia and zirconia were followed by an exponentially rapid growth of groups active in the area. Furthermore, this discovery has stimulated investigation of ferroelectricity in conventionally non-ferroelectric polar materials including $Mg_xZn_{1-x}O$, $Al_xB_{1-x}N$, and $Al_{1-x}Sc_xN$ suggesting that ferroelectricity can be ubiquitous in nitrides and oxides.[11-16]

From the first reports on ferroelectricity in hafnia, controversy emerged on the origins of the switchable spontaneous polarization in this material.[17] Early density-functional theory calculations demonstrated the feasibility of intrinsic polarization instability in these structures;[18-20] however, experimental studies have demonstrated that many aspects of ferroelectric behavior in hafnia were considerably complex compared to classical ferroelectrics. These include a more exaggerated wake-up effect, i.e. when classical ferroelectric hysteresis loops emerge after cycling of the fabricated devices.[21, 22] Similarly, ferroelectric behaviors in hafnia-based systems possess an anomalously strong dependence of electrode interfaces and preparation conditions.[23] Jointly, these observations suggest that ferroelectric-like behavior in these materials either originate or are mediated by alternative mechanisms.[24] The former can include the formation of mesoscale chemical dipoles and vacancy ordering, whereas the latter suggest the potential coupling between ferroelectric and ionic or strained subsystems. Additionally, a large discrepancy between nanoscale free surface and bulk capacitor-based measurements has been observed throughout the community, further suggesting complex mechanisms underpin hafnia's functionality.

Driven by these considerations, a number of groups have proposed that ferroelectricity in hafnia can be mediated by ionic defects, most notably oxygen vacancies.[22, 25] Furthermore, the polymorphism of hafnia was posed to be associated with multiple factors including strain, doping, field-induced phase transitions, and changes in oxygen stoichiometry resulting in complexities associated with stabilizing the ferroelectric orthorhombic phase.[26-28]

Likewise, scaling effects have been observed to increase the complexity of hafnia-based systems. Unlike conventional ferroelectrics (e.g. perovskites, such as $BaTiO_3$) where the polarization is known to diminish with reducing crystal size, in hafnia systems the stability of the non-equilibrium phases increases with decreasing crystal size. Several groups have experimentally shown a strong dependence of phases present with film thickness. For example, films that are thicker than approximately 15 nm often show increased phase fractions of the equilibrium, non-ferroelectric monoclinic phase[29, 30] and adding artificial interfaces to disrupt the lattice has been successful in stabilizing the non-equilibrium phase to larger thicknesses.[31] These observations point to a large surface/interface energy contribution to phase stability in the hafnia system, which has been well-known in fluorite oxides since early work on ZrO_2.[32] Computational efforts have corroborated this surface/interface effect on phase stability where different thickness regimes of metastable phase formation have has been shown to be 5, 15, and approximately 35 nm in HfO_2, $Hf_{0.5}Zr_{0.5}O_2$, and ZrO_2, respectively.[33] Furthermore, recent computational and experimental reports have shown there is, practically, no size limit for ferroelectricity in hafnia-base systems. Specifically, Zhao, *et al.*,[28] showed that the polar phase is stabilized by an antipolar mode that results in extreme size scaling and Cheema, *et al.*,[34] have shown experimental evidence for reversible polarization in films that are only 1 unit cell thick. Given that surfaces and interfaces play a large role in phase stability, changes of electrochemical boundary conditions will strongly impact a system that is stabilized by surface energy.

While discussing the state-of-the-art understanding of ferroelectricity in hafnia, we note that over the last decade anomalous ferroelectric-like behaviors were also observed by the classical ferroelectrics community. For example, observations of hysteresis loops and switchable remanent polarization in materials where polarization exists in the bulk but is expected to disappear in thin layers, or in materials that do not have polarization instability were reported.[35] The characteristic aspect of these systems is the presence of a continuum of polarization states (as opposed to binary polarization in classical ferroelectrics), and sensitivity to environmental conditions. Furthermore, often these phenomena could be generally observed on free surfaces via Piezoresponse Force Microscopy (PFM), but not device structures formed from these films.[36] Here, it was argued that the coupling between the bulk ferroelectric instability and surface electrochemistry[37, 38] can give rise to new ferroionic states.[39, 40] More generally, these studies have pointed to the possible key role of electrical-potential compensation phenomena at ferroelectric surfaces and interfaces beyond a classical dead-layer model.

Here, we argue that the observed ferroelectric behaviors in hafnia can be ascribed to the interplay between bulk antiferroelectric instability with non-linear surface (or interface) charge compensation, manifested in the bulk and at the interfaces, respectively. This fundamental disparity in the spatial localization of bulk ferroelectric instability and interfacial behaviors renders non-local polarization switching and phase evolution, resulting in complex time- and voltage-responses. To explore these assertions, we aim to control the ionic and electrostatic degrees of freedom separately. Note that while this separate control is highly challenging to achieve at the internal interfaces (simply because biasing affects both electrostatic and electrochemical potentials jointly), such control can be readily accomplished at the surfaces via the partial pressure of oxygen and temperature. As such, decoupling of electrochemical and electrostatic controls realized via environmental and ultra-high vacuum PFM, we show that these materials demonstrate a rich spectrum of ferroic behaviors including oxygen partial pressure- and temperature-induced transitions between FE and AFE behaviors.

Structural Phase Identification

As a model system to explore hafnia-based antiferroelectric instabilities, we have chosen a 17 nm $Hf_{0.5}Zr_{0.5}O_2$ (HZO) thin film grown via plasma-enhanced atomic layer deposition (ALD) on a silicon substrate with TaN bottom electrodes (see Methods Section). To evaluate the intrinsic material properties of HZO, we employ structural and chemical characterization as seen in Figure 1. The atomic force microscopy (AFM) topography reveals a surface morphology consisting of 25 nm average grain sizes (Figure 1a) and a root-mean-square roughness of 1.2 nm, consistent with previous reports of ALD prepared HZO.[41] Figure 1b shows the cross-section of HZO film (in the awoken state) using atomically resolved high-angle annular dark field STEM micrographs, where a single grain comprising the full thickness of the film is observed, together with the TaN bottom electrode and the top protective cap (Pt) used for sample preparation. Position averaged convergent electron beam diffraction (PACBED) was also acquired to facilitate the determination of the phases (Figure 1b). Although a lack of mirror plane should be observed across the dashed axis bisecting the pattern for the $Pca2_1$ polar phase, the experimental pattern shown in Figure 1b revealed a more symmetric distribution. These results together with the presences of weak reflections in the FFT revealed a coexistence of two orthorhombic phases, including *Pca*2_1 (polar) and *Pbca* (anti-polar) oriented along the [110] and [101] directions respectively, as shown in Figure 1b (right panel), establishing the presence of competing ferroelectric-antiferroelectric states.[42]

To identify the relative phase fractions within the HZO thin film, Figure 1c shows grazing incidence X-ray diffraction (GI-XRD) differentiating the monoclinic from orthorhombic phase finding an 80.9% orthorhombic phase. GI-XRD is performed because the grazing incidence angle is near the critical angle for the hafnia films. This, in turn, leads to an increased sampling volume and signal-to-noise ratio in the XRD scan. Importantly, both orthorhombic *Pbca* or the *Pca*2_1 phases are virtually indistinguishable from the tetragonal phase via XRD; thus, the corresponding peak is labeled as orthorhombic/tetragonal. Lastly, chemical homogeneity can strongly influence the ferroelectric properties of HZO. As seen from Figure 1d, a relatively uniform distribution of hafnium (green), zirconium (blue), and oxygen (red) is observed. Moreover, the average profiles shown in Figure 1e shows a relatively uniform z-profile of the individual constituents.

Environmentally dependent electromechanical response

To investigate the coupling between the bulk ferroelectric instability and surface electrochemistry, band excitation (BE) PFM measurements were performed in three different environments on pristine HZO, i.e., ambient, glove box (<10 ppm O_2) and in UHV vacuum (2×10^{-10} Torr) to control the effective partial pressure of oxygen. Note that in contrast to single frequency PFM, BE-PFM utilizes a non-sinusoidal signal with a defined band in frequency space to independently detect resonance frequency and response amplitude, ultimately mitigating topographic crosstalk.[43] Figure 2 shows classical box-in-a-box poling experiments acquired in all three environments using +7 V and -7 V DC biases to pole 600 nm and 300 nm areas, respectively (solid and dotted boxes, Figure 2a). Figure 2a,d displays the BE-PFM amplitude and phase, respectively, acquired after poling in ambient conditions. Here, the formation of domain walls and oppositely poled regions are unobserved. Interestingly, localized regions of the BE-PFM phase (Figure 2d) seem to vary across the region of interest suggesting the presence of additional mechanisms, such as nonuniform polarization or surface states. As the environmental oxygen partial pressure is reduced, i.e. glovebox measurements, localized areas with decreased amplitude are observed (Figure 2b,e) in the +7 V poled region. In contrast, as the environmental oxygen partial pressure is further reduced via UHV conditions, clear domain wall formation and polarization switching is observed within the poled regions (Figure 2c,f) with the presence of localized unswitched areas in the +7 V poled region, suggesting possible localized defect clusters or non-ferroelectric monoclinic phases.

Next, HZO hysteresis loops were measured via band excitation piezoresponse spectroscopy (BEPS) in ambient, glove box, and UHV environments at 300 K and the results are shown in Figure 3. Here, classical hysteresis loops measured via PFM experiments are presented in the off-field state, where the PFM signal is measured at the zero-bias state after the application of a short bias pulse. This experimental protocol is chosen to minimize the contribution of electrostatic forces in the measured signal.[44, 45] In the interpretation of these measurements, it is postulated that the domain formed below the tip does not appreciably relax on the transition to zero probe DC bias, i.e. the domain walls are strongly pinned.[46] However, this assumption is not necessarily justified for materials, such as ferroelectric relaxors, with highly mobile domain walls, or materials with competing order parameters such as antiferroelectrics. Hence, here we present both on-field and off-field measurements.

Figure 3a,d shows the on-field and off-field switching behavior in ambient conditions, respectively. Here, clear pinching of the on-field hysteresis loop close to 0 V is observed, indicative of antiferroelectric-like behavior. Notably, the pinched loop is slightly offset from 0 V suggesting the presence of a built-in potential. Examination of the off-field loop also shows antiferroelectric-like behavior, suggesting metastability of the bias-induced polar phase. However, in an argon environment, i.e. glovebox measurements, a lower degree of pinching accompanied with a narrowing of the hysteresis loop is observed in the on-field state with no remanent ferroelectric switching observed in the off-field state (Figure 3b,e and Figure S9), indicating a paraelectric-like phase. In contrast, in UHV conditions ($2x10^{-10}$ Torr) a widening of the on-field loop with the absence of pinching is observed, indicating the onset of the ferroelectric state (Figure 3c). Moreover, the off-field hysteresis loops show clear remanent switching (Figure 3f). Strikingly, a similar trend can be observed when cycling $Hf_{0.5}Zr_{0.5}O_2$ within a capacitor geometry (Supplementary Figure S8) where cycling facilitates a more traditional ferroelectric response (i.e. wake-up phenomena). In particular, scanning probe measurements performed in a reducing environment (i.e. in UHV) on a free surface produce similar ferroelectric responses as bulk measurements (i.e. capacitor geometries) post cycling, suggesting a common mechanism.

It is important to note, the properties of surface coupled ferroelectrics, i.e. ferroionics , and ferroelectric relaxors induced by the Vogel-Fulcher relaxation of dipoles will have multiple similarities in terms of relaxation, hysteresis loop shapes, etc. This is unsurprising, given that relaxors are governed by the bulk dipolar disorder, and ferroionic systems are coupled with the surface. That said, the key effect observed in Figure 3a-f is the environment-induced transition that clearly illustrates the role of surface electrochemical coupling. While relaxors are affected by surface states, the surface states are irrelevant to the emergence of the relaxor phase and only modifies it close to surfaces/interfaces. Moreover, the bulk measurements (Supplementary Information Figure S8) seem to rule out the role of the bulk disorder in the material, and hence relaxor-like behaviors.

In light of these observations, it is also important to consider how traditional ferroelectrics, e.g. $BaTiO_3$, behave in all three environments. As such, for comparison, average off-field hysteresis loops of 80 nm thick $BaTiO_3$ are shown in Figure 3g-i. Here, clear hysteretic switching can be observed in all environments, as expected; however, differences in the hysteresis loop shape are readily observed suggesting, in reducing environments (UHV), oxygen vacancies can migrate and/or be created within the PFM probing volume inducing unprecedented responses, similar to recently reported observations.[47] Importantly, environmentally induced surface effects, such as the propensity for surface adsorbates or oxygen vacancy formation at the surface, can have substantial impact on the ferroelectric behavior as seen in HZO.[48]

These observations suggest the electrochemical state of the surface strongly couples to the ferroic behavior in the hafnia system. In other words, as the environmental oxygen partial pressure and total pressure via UHV is reduced, the chemical potential driving force will reduce compensating mechanisms (e.g. surface absorbates or ionic species) and subsequently decrease the oxygen content at the surface, which is expected to increase the negative surface charge (i.e. compensated oxygen vacancies) potentially giving rise to the observed ferroelectric stability. As such, it is worth mentioning the placement of conductive electrodes on hafnia's surface can also provide the necessary interfacial charge to stabilize the ferroelectric behavior, congruent with previous reports.[49]

Phenomenological Landau-type modeling

To gain further insight, we apply a previously developed formalism for an antiferroelectric film coupled to surface electrochemical reactions.[50] This phenomenological formalism considers competing polar and structural anti-polar distortive long-range orders in $(Hf,Zr)O_2$ at the mesoscopic level with the possible role of doping and oxygen vacancies, in the presence of finite surface (or interface) density of ionic and electronic states (see Supplementary Information section A and B). We use this formalism to plot the phase diagram of a 20-nm thick HZO film as temperature versus relative oxygen pressure (Fig. 4a) where the dashed curves separate different type of loops corresponding to the nonpolar antiferroelectric-like ferroionic phase (AFEI), the intermediate paraelectric-like phase (PE), and the ferroelectric-like ferroionic phase (FEI). This classification is valid for quasi-static changes of the applied voltage and is based on the shape of polarization hysteresis loops, which continuously transforms from a double AFE-type loop through hysteresis-less PE curve towards a single FE-type loop as shown in Fig. 4a. Importantly, the experimentally measured hysteresis loops acquired as a function of relative oxygen partial pressure are in great qualitative agreement with the phenomenological Landau-type modeled loops, suggesting the experimentally observed behavior is due to the proposed AFEI-PE-FEI transition mechanisms, i.e., the interplay between bulk antiferroelectric-instability with non-linear surface charge compensation. Also, it is important to consider that the modeled behavior indicates the AFE-type loops transform to FE-type with the increase of the applied voltage frequency (Fig.s S1-S7 in the Supplement Information), which is potentially important for hafnia-based applications.

A critical component of the modeled behavior is the dynamic electric field dependencies on the out-of-plane polarization (black curves), and positive (red curves) and negative (blue curves) surface charges, which are calculated for decreasing oxygen pressure and different relaxation times between the positive (τ_p) and negative (τ_n) screening charges, $\tau_n \ll \tau_p$ (Fig. 4b-d). A clear increase in hysteresis loop opening, or ferroelectric-like behavior, can be seen with decreasing relative oxygen partial pressure (Fig. 4b-d), which is accompanied by an increase in negative surface charges (potentially compensated oxygen vacancies) and relatively constant positive surface charges. Congruently, the corresponding normalized dielectric susceptibilities χ/χ_o are shown in Fig. 4e-g (calculation parameters selected based on experimental observations are listed in Table S1 in Supplementary Information). The susceptibility, corresponding to FEI-type states, has two peaks standing on a wide high "base". The frequency behavior of the χ/χ_o in the

FEI state is different from those in FE and PE states. Indeed, χ/χ_o in the FE state has two sharp peaks at coercive field (without any base), which remain unchanged in a wide frequency range; and χ_e in the PE state has a frequency-sensitive central maximum without any side peaks. In the FEI state, the decrease of oxygen pressure leads to the continuous decrease of the base height, its horizonal shift, and increase of the peaks' height. Note that the features of polarization loops, characteristic to AFEI states, can be seen at much lower frequencies of applied bias (compare Fig. S2 and S3, with S4 and S5 in Supplementary Information).

Ferroionic temperature dynamics

We further proceed to explore temperature dynamics in an effort to probe the complex HZO ionic-electrostatic mediated phase diagram. Here, we note that in classical ferroelectrics the hysteresis loops are weakly dependent on temperature away from the phase transition, and generally expected to remain symmetric. At the same time, ferroionic systems were predicted to give rise to a broad range of temperature-mediated behaviors, including the transitions between ferroelectric-like and antiferroelectric-like behaviors.

Accordingly, we collected temperature dependent BEPS hysteresis loops under ambient and UHV conditions (Figure 5) to vertically transverse the predicted complex phase diagram in a high and low oxygen partial pressure environment, respectively. Figure 5a-c shows hysteresis loops acquired at 360, 420, and 500 K in ambient with the on-field and off-field shown in the left and right panels. At 360 K (Figure 5a), a clear pinching of the hysteresis loop is still observed in the on-field state, indicating anti-ferroelectric like behavior; however, the off-field begins to lose the metastable bias-induced polar phase observed in Figure 3d, i.e. the loop opening with negative bias. Furthermore, as the temperature is increased to 500 K, the pinched hysteresis loops begin to collapse, similar to Figure 3b,e, suggesting the onset of a paraelectric-like phase transition. Jointly, these observations imply HZO begins to undergo an antiferroelectric to paraelectric phase transition at elevated temperatures (>500 K) under ambient conditions, in great agreement with the predicted complex phase diagram in Figure 4a.

Lastly, Figure 5d-i shows average on-field, off-field, and on-field differential hysteresis loops acquired at 108, 360, and 420 K from a 1 μm area (10x10 grid) in UHV. At 108 K, the on-field loops (Figure 5d, left panel) display an observable pinching near 0 V while the off-field (Figure 5d, right panel) loops have an asymmetric response, particularly along the piezoresponse axis, similar to Figure 4i-l. Interestingly, when evaluating the on-field differential dP/dV (Figure 5g), two clear kinks are observed corresponding to pinching of the on-field loop (dotted circles Figure 5g-i). Here, we identify the first order changes in dP/dV to be associated with antiferroelectric behavior. Upon increasing the temperature to 360 K, no pinching was observed, which is readily apparent from the dP/dV loops (Figure 5h). Furthermore, the off-field loops at 360 K have distinctly different switching behavior; namely, a decrease in the coercive field relative to hysteresis loops acquired at 108 K, which is in remarkably good agreement with the predicted behavior shown in Figure 4a. Lastly, at 420 K, the on-field loop pinching (Figure 5f, left panel) reappears, which is confirmed via kinks in the dP/dV spectra (Figure 5i), suggesting a temperature-induced ferroelectric-antiferroelectric instability. Exploring the off-field response at 420 K reveals symmetric hysteresis loops with a larger coercive voltage, potentially originating from mechanisms such as ionic motion, which is certainly plausible at elevated temperatures as observed in other material systems.[51, 52] Overall, the temperature dependent BEPS measurements are in qualitative agreement with the modeled results with the exception of observations at higher temperatures, i.e. at 420 K antiferroelectric-like behavior is observed suggesting the presence of additional coupled mechanics (e.g. ionic motion).

Outlook

To summarize, here we reveal strong dependence of the ferroelectricity in hafnia on the external environment, explored over a range of partial pressures and temperatures. These observations confirm that electrochemical state of the surface strongly couples to the ferroelectric phase transition and stability in these materials and is inseparable from ferroelectricity. These observations can be described in terms of the antiferroionic model of coupled antiferroelectric bulk and surface electrochemical phenomena. Importantly, the free energies of the various hafnia phases are compositionally-dependent and the surface energy term will impact them each differently. As such, we expect the observed behavior of the ferroionic response would hold for any ferroelectric fluorite, but the phase diagrams will differ depending on composition (e.g. the pO_2 and temperature boundaries may differ).

Overall, these mechanisms can provide a general explanation for the range of the unusual phenomena obtained in hafnia and similar binary oxides, including slow evolution of the ferroelectric properties, dependence on processing conditions, and the observed discrepancy between ferroelectric measurements acquired on a free surface and in a capacitor geometry. This, in turn, creates novel opportunities for the discovery and optimization of these material systems, such as circumventing the wake-up effect, which is enabled via independent ionic control.

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Author contributions

K.P.K performed and conceived experiments, (A.N.M and E.A.E) performed theoretical calculations, Y.L. performed ambient piezoresponse force microscopy measurements, (S.T.J., S.F., T.M. and J.F.I.) grew samples and performed bulk characterization, (S.C. and E.D.) performed scanning electron transmission microscopy and analysis, (K.P.K. and S.V.K) conceived the project. All authors contributed to discussions and the final manuscript.

Competing interests

The authors declare no competing interests.

Figure 1. Structural and chemical characterization of $Hf_{0.5}Zr_{0.5}O_2$. (a) Surface morphology of $Hf_{0.5}Zr_{0.5}O_2$ acquired via atomic force microscopy. (b) STEM high-angle annular dark field micrograph (left panel) with magnified region (right panel) showing a grain with coexisting orthorhombic $Pca2_1$ and $Pbca$ phases imaged along the [110] and [101] zone axis, respectively. Top inset (right panel) presents the experimental FFT showing the presence of two orthorhombic phases. Right inset (right panel) shows position averaged convergent electron beam diffraction representing experimental (top), modelled $Pca2_1$ phase (middle), and modelled $Pbca$ phase (bottom). (c) Grazing incidence X-ray diffraction identifying orthorhombic/tetragonal and monoclinic phases. Note, orthorhombic/tetragonal phase accounts for 80.9% of structural phase. (d) Energy-dispersive X-ray spectroscopy maps. (e) Averaged EDS profiles as a function of distance from the top electrode acquired from EDS map (boxed region in panel d) for each element.

Figure 2. Band excitation piezoresponse force microscopy ferroelectric poling. $Hf_{0.5}Zr_{0.5}O_2$ thin films poled via AFM bias of +7 V (600 nm solid box, panel **(a)**) and -7 V (300 nm dotted box, panel **(a)**) in three different environments: **(a,d)** ambient, **(b,e)** argon, and **(c,f)** ultra-high vacuum (amplitude and phase response respectively). Note, images are acquired immediately after poling via bias applied to AFM cantilever.

Figure 3. Band excitation piezoresponse spectroscopy hysteresis loops. Average on-field and off-field hysteresis loops of $Hf_{0.5}Zr_{0.5}O_2$ over 2 cycles acquired from a 10x10 grid over a 1 μm area in three different environments: **(a,d)** ambient, **(b,e)** argon, and **(c,f)** ultra-high vacuum. For comparison, average off-field hysteresis loops of 80 nm thick $BaTiO_3$ are shown in panels **(g-i)** acquired in ambient, argon, and ultra-high vacuum, respectively. Top **(a-c)** and bottom **(d-i)** panels correspond to on-field and off-field hysteresis loop, respectively. All data is presented as mean values +/- standard deviation.

Figure 4. Phenomenological Landau-type modeling. (a) Phenomenological Landau-type modeling of HZO temperature versus relative oxygen pressure phase diagram. The thickness of HZO film is $h = 20$ nm. Dashed curves separate different type of loops corresponding to the nonpolar AFEI-phase, the intermediate PE-like phase and the FE-like FEI phase. **(b)-(d)** Normalized polarization (P/P_s) versus normalized dynamic electric field (E/E_c) for the out-of-plane polarization (black curves), positive (red curves) and negatives (blue curves) surface charges calculated for decreasing oxygen pressure (10^{-2}, 10^{-4} and 10^{-10} pO_2) and different relaxation times of the screening charges, $\tau_n = 10^{-2}\tau_p$. **(e)-(g)** Corresponding electric field dependences of the normalized dielectric susceptibility (χ/χ_o). Polarization, electric field, and dielectric susceptibility are normalized by spontaneous polarization (P_s), coercive field (E_c), and zero electric field susceptibility (χ_o), respectively. Parameters used in calculations are listed in **Table S1.** Note that plots (b)-(g) are calculated for smaller amplitude of applied bias than the polarization loops shown in the diagram (a). Comparison with Figure S2 shows that the quasi-static and dynamic loops look different.

Figure 5. Temperature dependent band excitation spectroscopy. Average band excitation piezoresponse spectroscopy hysteresis loops acquired over a 10x10 grid, 1 µm area, and 3 cycles. Measurements acquired at **(a)** 360 K, **(b)** 420 K, **(c)** 500 K in ambient and **(d)** 108 K, **(e)** 360 K, **(f)** 420 K in ultra-high vacuum. **(g-i)** Differential on-field dP/dV hysteresis loops are derived from **(d-f)**, respectively. Note, dotted circles in panel **(g)** and **(i)** highlight kinks associated with antiferroelectric behavior. Left and right panels in **(a-f)** correspond to on-field and off-field, respectively.

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Methods

Material Synthesis

Metal-Insulator-Metal (MIM) TaN/HZO/TaN devices were prepared on 500 μm thick silicon substrates. The planar bottom TaN electrode was deposited through a DC sputtering process from a sintered TaN target to a thickness of 100 nm at room temperature. Next, an Oxford FlexAL II Plasma-Enhanced Atomic Layer Deposition system was utilized to grow a $Hf_{0.5}Zr_{0.5}O_2$ film to a thickness of 17 nm at 260 °C using tetrakis(ethylmethylamido)hafnium (TEMA Hf) and tetrakis(ethylmethylamido)zirconium (TEMA Zr) as HfO_2 and ZrO_2 precursors, respectively, in 5:5 super cycle ratios to control film composition. An oxygen plasma was employed as the oxidant. Following HZO deposition, a 20 nm thick planar TaN top electrode was prepared through an identical process to the bottom electrode, and the film stack was annealed within an Allwin21 AccuThermo 610 Rapid Thermal Processor for 30 seconds under a N_2 atmosphere. For the sample for piezoresponse force microscopy measurements, the entire film was exposed to a SC-1 (5:1:1 H_2O:30% H_2O_2 in H_2O:30% NH_4OH in H_2O) etch solution at 60 °C for 45 minutes to fully remove the top electrode layer. To prepare electrodes, 50 nm thick Pd electrodes were deposited through a shadow mask and the exposed TaN was removed using the SC-1 treatment. The discrete Pd pads served as a hard mask.

Polarization Measurements

Polarization versus electric field measurements were performed with a Radiant Technologies Precision LC II ferroelectric tester with a measurement period of 1 ms. An Instec hot-stage probe station equipped with a turbo molecular pump was used to measure performance at in ambient and high vacuum conditions at 300 K, 360 K, and 420 K.

Atomic Force Microscopy Measurements

All piezoresponse force microscopy measurements were captured using Budget Sensor Multi75E-G Cr/Pt coated AFM probes (~3 N/m). Band excitation was achieved by coupling the AFMs with an arbitrary wave generator and data acquisition electronics based on a National Instruments fast DAQ card. Custom software was used to generate the probing signal and store local BE and hysteresis loops. All band excitation PFM measurements were collected using frequencies ranging from 300-400 kHz, with subsequent simple harmonic oscillator fits applied to the collected spectra to extract amplitude and phase at the resonance frequency. Ambient PFM measurements were taken using an Oxford Instruments Cypher atomic force microscope. Glovebox measurements were taken using a Bruker Dimension Icon fitted inside a MBRAUN MB200B series glovebox. Ultra-high vacuum measurements were deployed via a modified ultra-high vacuum Omicron AFM-STM microscope controlled from a Nanonis real time controller at a pressure of ~2×10^{-10} Torr. Note, for all piezoresponse force microscopy measurements, the same pristine HZO sample was used. However, additional measurements on multiple samples highlighting reproducibility can be found in the Supplementary Information (Figure S11). For all post poling images (Figure 2), images were acquired directly after writing with a PFM tip bias of ±7V.

Scanning Transmission Electron Microscopy Measurements

STEM images were acquired in a ThermoFisher Themis operated at 200kV, using a high angle annular dark field (HAADF), a probe convergence angle of 17.9 mrads and a camera length that resulted in acceptance angles between and 60 – 200 mrads. A set of 20 frames were collected using fast-dwell-time (200ns) to avoid drift that can cause distortions in the images. The frames were later registered using cross-correlation to improve the signal-to-noise ratio. Energy dispersive X-ray spectroscopy was used to acquire iterative elemental maps of 512 x 512 pixels with a dwell time per pixel of 10 μs at 200 keV during 10 min using a Super-X EDS detector. The maps were constructed using N K_{α}, O K_{α}, Zr K_{α}, Hf L_{α}, Ta L_{α} and Pt L_{α} lines. Position-averaged convergent beam electron diffraction patterns (PACBED) were formed by averaging convergent beam electron diffraction patterns while scanning the sample with the STEM probe, using a convergence semi-angle of 17.9 mrads and 500 ns of dwell-time. Patterns were collected during 1 s over the scanned area using a Ceta camera from Thermo Scientific. Position-averaged convergent beam electron diffraction patterns (PACBED) were calculated by multislice computer simulations using Dr. Probe V1.9 software package [https://doi.org/10.1016/j.ultramic.2018.06.003], and compared with experimental patterns to determine the phases and estimate samples thickness.

Theoretical Modeling and Analysis

Data supporting theoretical results was visualized in Mathematica 12.2 [https://www.wolfram.com/mathematica] and can be found at the Notebook Archive (https://notebookarchive.org/xxxx). Thresholding for coercive field and remanent polarization maps were acquired using a ±1V and ±0.2 a.u. threshold, respectively.

Data Availability

The data that support the findings of this study are available from the corresponding author upon request.

Code Availability

Detailed information related to the codes used in this manuscript are available in the Supplementary Information files or from the corresponding authors upon request.